Extending the Oracle Approximating Shrinkage to General Distributions

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***Abstract*—The Oracle Approximating Shrinkage (OAS) estimator provides a closed-form, data-driven shrinkage intensity for regularizing sample covariance matrices toward a scaled identity target, but its derivation assumes Gaussian data. In practice, financial returns, neural signals, and many other high-dimensional datasets exhibit heavy tails and non-zero kurtosis, violating this assumption. We extend OAS to any distribution with finite fourth moments by incorporating a correction term α built from the fourth-order joint cumulants of the data. The correction vanishes under Gaussianity, recovering the OAS formula as a special case; it admits a consistent $O(np)$ sample estimator and satisfies $\rho \in [0, 1]$ exactly for both the population cumulant and its sample analogue. Simulations and an equity-returns study confirm that the correction closes the gap between OAS and the Ledoit–Wolf estimator under non-Gaussian data, with OAS-Extended performing comparably across all tested settings. The contribution is a principled correction for OAS practitioners facing heavy-tailed data; structurally, it is distinct from prior work extending Ledoit–Wolf to elliptical distributions, which optimizes against a fixed oracle target and yields a different closed-form formula.**

***Index Terms*—covariance matrix estimation, shrinkage estimation, Oracle Approximating Shrinkage, fourth-order cumulants, non-Gaussian distributions, heavy tails, high-dimensional statistics, portfolio optimization**

I. Introduction

Estimating a covariance matrix from data is a fundamental task across a wide range of disciplines, including portfolio optimization [2], signal processing [1], and neuroscience. In the classical low-dimensional regime, the sample covariance matrix $\hat{S}$ is a consistent and unbiased estimator of the true covariance Σ. However, in the modern high-dimensional setting where the number of variables p is comparable to or exceeds the number of observations n, $\hat{S}$ becomes poorly conditioned or singular, and its eigenvalues are known to be severely dispersed relative to those of Σ.

Shrinkage estimators address this by forming a convex combination

$$\hat{\Sigma} = (1 - \rho)\hat{S} + \rho\hat{F},$$

where $\hat{F}$ is a well-conditioned target matrix and $\rho \in [0, 1]$ is a shrinkage intensity. The celebrated Ledoit–Wolf (LW) estimator [2] chooses ρ to minimize the expected Frobenius loss $E[\|\hat{\Sigma} - \Sigma\|_F^2]$ against the oracle target $\hat{F} = (\mathrm{Tr}(\Sigma)/p)\, I$. The Oracle Approximating Shrinkage (OAS) estimator of Chen et al. [1] takes a different approach: it uses the data-dependent target $\hat{F} = (\mathrm{Tr}(\hat{S})/p)\, I$ and derives ρ via a fixed-point iteration that converges to a closed-form expression. This distinction — random versus fixed target — leads to structurally different formulas and different finite-sample behavior, with OAS typically outperforming LW at small n/p ratios under Gaussian data.

Both estimators were originally derived under the assumption of multivariate Gaussian data. In many practical settings this assumption fails. Financial asset returns are well-documented to exhibit excess kurtosis and heavy tails. More broadly, any distribution in the elliptical family — including the multivariate Student-t — introduces fourth-order cumulant structure that the Gaussian-derived formulas ignore. Under such distributions, applying the OAS formula amounts to a misspecified estimator, and the resulting shrinkage intensity is suboptimal.

The key insight is that the Gaussian derivation of OAS depends on two expectations — $E[\mathrm{Tr}(\hat{S}^2)]$ and $E[\mathrm{Tr}^2(\hat{S})]$ — each of which acquires an additional term α/n under a general distribution, where $\alpha = \sum_{i,j} \kappa(X_i, X_i, X_j, X_j)$ aggregates the fourth-order joint cumulants of the data. Incorporating this term propagates through the fixed-point iteration and yields a corrected closed-form shrinkage intensity that reduces to the OAS formula when $\alpha = 0$. We also provide a consistent sample estimator $\hat{\alpha}$ and prove that the corrected formula preserves $\rho \in [0, 1]$ for both the population cumulant and the sample estimator, so an OAS practitioner facing non-Gaussian data can adopt the extended formula without changing frameworks.

II. Related Work

The closest prior work is the extension of LW shrinkage to elliptical distributions by Ollila and collaborators [3], [4], [5], which minimizes

$$E\Big[\|(1 - \rho)\hat{S} + \rho\mu I - \Sigma\|_F^2\Big]$$

where $\mu = \mathrm{Tr}(\Sigma)/p$ is a *fixed* oracle scalar, yielding a closed-form expression parameterized by the scalar elliptical kurtosis κ and the sphericity $\gamma = p\,\mathrm{Tr}(\Sigma^2)/\,\mathrm{Tr}^2(\Sigma)$.

The OAS framework of [1] differs structurally: the target $\hat{F} = (\mathrm{Tr}(\hat{S})/p)\, I$ is *random and data-dependent* rather than the oracle μI. This changes the optimization problem and produces a different formula via fixed-point iteration; the two approaches are not equivalent even in the Gaussian case. Consequently, the elliptical results of [3], [4] do not imply an extension of OAS, and to the best of our knowledge the present work is the first to provide one.

Our result is also more general in distributional scope. The Ollila line restricts to elliptical distributions, for which $\kappa(X_i, X_i, X_j, X_j) = \kappa(\Sigma_{ii}\Sigma_{jj} + 2\Sigma_{ij}^2)$ for a scalar κ. Our

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result holds for any distribution with finite fourth moments, where α need not have this constrained form; elliptical distributions are a special case.

III. Corrected Oracle Intensity

a) Setup and notation.: Let $X^1, \dots, X^n \in \mathbb{R}^p$ be i.i.d. draws from a centered distribution with covariance Σ and finite fourth moments. The sample covariance is

$$\hat{S} = \frac{1}{n} \sum_{k=1}^{n} X^k (X^k)^T,$$

and the shrinkage target is $\hat{F} = \frac{\mathrm{Tr}(\hat{S})}{p} I$. The setup and following equations follow [1]:

$$\rho_O = \frac{E[\mathrm{Tr}((\Sigma - \hat{S})(\hat{F} - \hat{S}))]}{E[\|\hat{S} - \hat{F}\|_F^2]}, \tag{1}$$

$$E[\|\hat{S} - \hat{F}\|_F^2] = E[\mathrm{Tr}(\hat{S}^2)] - \frac{E[\mathrm{Tr}^2(\hat{S})]}{p}. \tag{2}$$

Here we depart from [1]: rather than assuming Gaussianity, we allow the distribution to be arbitrary with finite fourth moments. Observe that

$$\begin{aligned} E[\mathrm{Tr}(\hat{S}^2)] &= \sum_{i,j} \frac{1}{n^2} \sum_{k,l} E[X_i^k X_j^k X_i^l X_j^l] \\ &= \frac{n-1}{n} \sum_{i,j} \Sigma_{ij}^2 + \frac{1}{n} \sum_{i,j} E[X_i^2 X_j^2] \\ &= \frac{n-1}{n} \mathrm{Tr}(\Sigma^2) + \frac{1}{n} \sum_{i,j} E[X_i^2 X_j^2]. \end{aligned} \tag{3}$$

Similarly,

$$\begin{aligned} E[\mathrm{Tr}^2(\hat{S})] &= \sum_{i,j} \frac{1}{n^2} \sum_{k,l} E[(X_i^k)^2 (X_j^l)^2] \\ &= \frac{n-1}{n} \mathrm{Tr}^2(\Sigma) + \frac{1}{n} \sum_{i,j} E[X_i^2 X_j^2]. \end{aligned} \tag{4}$$

The term $\sum_{i,j} E[X_i^2 X_j^2]$ appears in both expansions. For a centered distribution with finite fourth moments,

$$E[X_i^2 X_j^2] = \Sigma_{ii}\Sigma_{jj} + 2\Sigma_{ij}^2 + \kappa(X_i, X_i, X_j, X_j), \tag{5}$$

where $\kappa(X_i, X_i, X_j, X_j)$ is the fourth-order joint cumulant, vanishing under Gaussianity. Letting $\alpha = \sum_{i,j} \kappa(X_i, X_i, X_j, X_j)$,

$$\sum_{i,j} E[X_i^2 X_j^2] = \mathrm{Tr}^2(\Sigma) + 2\,\mathrm{Tr}(\Sigma^2) + \alpha. \tag{6}$$

Substituting (6) into (3) and (4),

$$E[\mathrm{Tr}(\hat{S}^2)] = \tfrac{n+1}{n} \mathrm{Tr}(\Sigma^2) + \tfrac{1}{n} \mathrm{Tr}^2(\Sigma) + \tfrac{\alpha}{n}, \tag{7}$$

$$E[\mathrm{Tr}^2(\hat{S})] = \mathrm{Tr}^2(\Sigma) + \tfrac{2}{n} \mathrm{Tr}(\Sigma^2) + \tfrac{\alpha}{n}, \tag{8}$$

matching the Gaussian results of [1] up to the α/n term. Substituting into (1) and simplifying yields the corrected oracle intensity:

$$\rho_O = \frac{(1 - \frac{2}{p}) \mathrm{Tr}(\Sigma^2) + \mathrm{Tr}^2(\Sigma) + \alpha(1 - \frac{1}{p})}{(n + 1 - \frac{2}{p}) \mathrm{Tr}(\Sigma^2) + (1 - \frac{n}{p}) \mathrm{Tr}^2(\Sigma) + \alpha(1 - \frac{1}{p})}. \tag{9}$$

IV. OAS-Extended

We now apply the OAS fixed-point framework of [1] to the corrected oracle formula (9). As in [1], we have

$$\hat{\rho}_{j+1} = \frac{(1 - \frac{2}{p}) \mathrm{Tr}(\hat{\Sigma}_j \hat{S}) + \mathrm{Tr}^2(\hat{\Sigma}_j) + \alpha(1 - \frac{1}{p})}{(n + 1 - \frac{2}{p}) \mathrm{Tr}(\hat{\Sigma}_j \hat{S}) + (1 - \frac{n}{p}) \mathrm{Tr}^2(\hat{\Sigma}_j) + \alpha(1 - \frac{1}{p})}, \tag{10}$$

$$\hat{\Sigma}_{j+1} = (1 - \hat{\rho}_{j+1}) \hat{S} + \hat{\rho}_{j+1} \hat{F}, \tag{11}$$

which replaces $\mathrm{Tr}(\Sigma)$ and $\mathrm{Tr}(\Sigma^2)$ with $\mathrm{Tr}(\hat{\Sigma}_j)$ and $\mathrm{Tr}(\hat{\Sigma}_j \hat{S})$. Let

$$\hat{\phi} = \frac{\mathrm{Tr}(\hat{S}^2) - \mathrm{Tr}^2(\hat{S})/p}{M}, \quad M = (1 - \tfrac{2}{p}) \mathrm{Tr}(\hat{S}^2) + \mathrm{Tr}^2(\hat{S}), \tag{12}$$

and $\beta = \alpha(1 - 1/p)/M$. Substituting (11) into (10) and simplifying,

$$\hat{\rho}_{j+1} = \frac{1 + \beta - (1 - \frac{2}{p}) \hat{\phi} \hat{\rho}_j}{1 + \beta + n\hat{\phi} - (n + 1 - \frac{2}{p}) \hat{\phi} \hat{\rho}_j}. \tag{13}$$

The change of variables

$$b_j = \frac{1}{1 + \beta - (n + 1 - \frac{2}{p}) \hat{\phi} \hat{\rho}_j} \tag{14}$$

reduces this to the linear recurrence

$$b_{j+1} = \frac{n\hat{\phi}}{1 + \beta - (1 - \frac{2}{p}) \hat{\phi}}\, b_j + \frac{1}{1 + \beta - (1 - \frac{2}{p}) \hat{\phi}}. \tag{15}$$

Its limit gives

$$\rho_{\text{OAS-E}} = \min\left(1,\, \frac{1 + \beta}{(n + 1 - \frac{2}{p})\, \hat{\phi}}\right), \tag{16}$$

the OAS-Extended estimator, recovering standard OAS when $\beta = 0$.

a) Sample estimator for α.: The natural moment estimator simplifies to

$$\hat{\alpha} = \widehat{\mathrm{Var}}(\|X\|^2) - 2\,\mathrm{Tr}(\hat{S}^2), \tag{17}$$

where $\widehat{\mathrm{Var}}(\|X\|^2) = \frac{1}{n} \sum_k \|X^k\|^4 - \mathrm{Tr}^2(\hat{S})$ is the sample variance of squared norms, convergent to α by the law of large numbers. This has a natural interpretation: $\hat{\alpha}$ measures the excess squared-norm variance beyond what a Gaussian with covariance $\hat{S}$ would produce ($2\,\mathrm{Tr}(\hat{S}^2)$ being the Gaussian value), and is computable in $O(np)$ additional time once $\hat{S}$ is formed. Appendix A shows that $1 + \hat{\beta} \geq 0$ holds exactly for every finite sample, so $\rho_{\text{OAS-E}} \in [0, 1]$ without clamping.

V. Simulation Study

We evaluate OAS-Extended against OAS [1], RBLW [1], LW [2], and the sample covariance. All experiments use $p = 120$ with AR(1) covariance $\Sigma_{ij} = 0.5^{|i-j|}$, $n/p \in \{0.25, 0.5, 1, 2, 5\}$, and 500 independent trials per setting. The smallest ratio $n/p = 0.25$ corresponds to $n = 30$ observations, ensuring the sample is large enough for stable estimation of $\hat{\alpha}$ while remaining well into the high-dimensional regime where $\hat{S}$ is singular. Performance is measured by normalized Frobenius loss $\|\hat{\Sigma} - \Sigma\|_F / \|\Sigma\|_F$.

Three distributions are tested: Gaussian ($\alpha = 0$) as a sanity check; multivariate Student-t with $\nu = 10$ degrees of freedom (excess kurtosis $= 1$); and $\nu = 5$ (excess kurtosis $= 6$), both scaled so $\mathrm{Cov}(X) = \Sigma$ exactly. For a multivariate Student-t with ν degrees of freedom, $\alpha = \frac{2}{\nu-4}[\mathrm{Tr}^2(\Sigma) + 2\,\mathrm{Tr}(\Sigma^2)]$, which grows rapidly as $\nu \to 4$, explaining why the correction is substantially larger under $t(\nu{=}5)$ than $t(\nu{=}10)$.

Table I reports mean losses. Under Gaussian data all four shrinkage estimators perform comparably, confirming no bias when $\alpha = 0$. Under non-Gaussian data four findings stand out. First, OAS-Extended recovers the performance lost by OAS's Gaussian misspecification, performing comparably to LW across all settings. Second, both dominate OAS substantially under $t(\nu = 5)$: losses fall by 46–48% at $n/p \leq 0.5$, $\sim$36% at $n/p = 1$, $\sim$24% at $n/p = 2$, and $\sim$16% at $n/p = 5$. Third, RBLW performs comparably to OAS under non-Gaussian data, confirming that neither the Rao-Blackwell construction of RBLW nor the fixed-point iteration of OAS provides a non-Gaussian correction. Fourth, the improvement of OAS-Extended over OAS grows with kurtosis and shrinks as n/p increases, exactly as predicted by the theory. Figure 1 shows loss curves and relative improvement over the full range $n/p \in [0.25, 10]$.

TABLE I
MEAN NORMALIZED FROBENIUS LOSS ($p = 120$, 500 TRIALS). LOWER IS BETTER.

n/p	Distribution	S	LW	RBLW	OAS	OAS-E
0.25	Gaussian	1.560	0.588	0.588	0.587	0.594
	$t(\nu{=}10)$	1.801	0.605	0.750	0.728	0.612
	$t(\nu{=}5)$	2.180	0.631	1.220	1.191	0.643
0.5	Gaussian	1.104	0.549	0.549	0.548	0.549
	$t(\nu{=}10)$	1.265	0.568	0.641	0.635	0.569
	$t(\nu{=}5)$	1.784	0.608	1.183	1.174	0.611
1.0	Gaussian	0.781	0.491	0.491	0.491	0.491
	$t(\nu{=}10)$	0.896	0.518	0.551	0.550	0.518
	$t(\nu{=}5)$	1.246	0.562	0.885	0.882	0.563
2.0	Gaussian	0.553	0.416	0.416	0.416	0.416
	$t(\nu{=}10)$	0.640	0.449	0.466	0.465	0.450
	$t(\nu{=}5)$	0.861	0.503	0.665	0.665	0.503
5.0	Gaussian	0.350	0.306	0.306	0.306	0.306
	$t(\nu{=}10)$	0.404	0.340	0.345	0.345	0.340
	$t(\nu{=}5)$	0.566	0.412	0.488	0.488	0.412

VI. EMPIRICAL STUDY: FINANCIAL RETURNS

We complement the Monte Carlo evidence with an application to daily equity returns. The universe consists of $p = 50$ large-cap US equities drawn from the S&P 500 index, using adjusted daily log-returns from January 2010 to December 2023 ($T = 3{,}521$ days) downloaded via `yfinance`, demeaned within each estimation window.

We adopt a rolling-window protocol with $T_{\text{in}} = 252$ days in-sample (approximately one trading year, giving $n/p \approx 5$) and a 21-day out-of-sample horizon stepped monthly, yielding $N = 155$ evaluation windows. At each step we estimate the covariance matrix with each estimator and form the global minimum variance (GMV) portfolio $\hat{w} \propto \hat{\Sigma}^{-1}\mathbf{1}$. Performance is measured by mean annualized realized volatility.

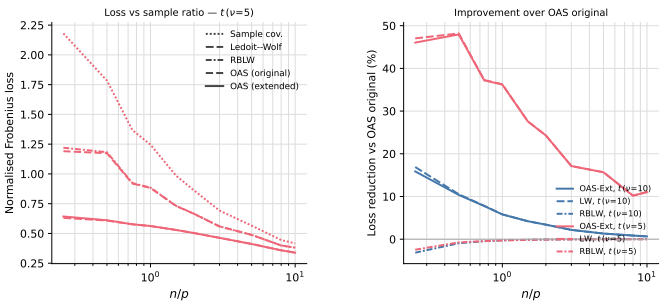


Fig. 1. Left: normalized Frobenius loss vs. n/p for $t(\nu{=}5)$ data ($p = 120$), all five estimators (dotted = sample cov., long-dash = LW, dash-dot = RBLW, dashed = OAS, solid = OAS-Extended). Right: relative loss reduction vs. OAS (%) for OAS-Extended (solid), LW (long-dash), and RBLW (dash-dot), blue = $t(\nu{=}10)$, red = $t(\nu{=}5)$.

$\hat{\alpha}$ is positive in every window (median 9.0×10^{-5}, mean 6.5×10^{-4}), confirming the heavy-tailed character of equity returns. The largest values coincide with periods of elevated market stress; the spike in early 2020 corresponds to the COVID-19 market dislocation.

Table II reports realized volatility. OAS-Extended reduces volatility by 2.0% relative to OAS (13.28% vs. 13.55%), approaching LW (13.276%), with improvement highly significant ($t = 3.83$, $p < 0.001$; bootstrap 95% CI for the mean variance reduction: $[2.0, 5.0] \times 10^{-6}$). RBLW closely tracks OAS ($t = 3.88$, $p < 0.001$), but its mean variance reduction is more than 30 times smaller than OAS-Extended's and the bootstrap CI rounds to zero at six decimal places — statistically detectable but economically negligible, confirming that neither RBLW nor OAS provides a meaningful non-Gaussian correction.

Table III breaks results down by market regime (VIX threshold = 20; 99 calm windows, 56 stress windows). In calm windows the correction yields a 1.0% vol reduction. In stress windows this rises to 4.0%, approximately four times larger. This concentration of the gain in high-$\hat{\alpha}$ periods is the direct empirical counterpart of the theory: when heavy-tailed behavior is most pronounced, the correction to the shrinkage intensity matters most.

TABLE II
MEAN ANNUALIZED REALIZED GMV VOLATILITY (%), $p = 50$, $N = 155$ WINDOWS. LOWER IS BETTER.

	S	LW	RBLW	OAS	OAS-E
All ($N = 155$)	13.972	13.276	13.551	13.552	13.282

TABLE III
MEAN ANNUALIZED REALIZED GMV VOLATILITY (%) BY MARKET REGIME (VIX THRESHOLD = 20). VALUES IN PARENTHESES ARE % REDUCTIONS VS. OAS.

Regime	N	OAS	LW	RBLW	OAS-E
Calm	99	13.785	13.645 (+1.0)	13.784 (+0.0)	13.649 (+1.0)
Stress	56	13.131	12.596 (+4.1)	13.129 (+0.0)	12.607 (+4.0)

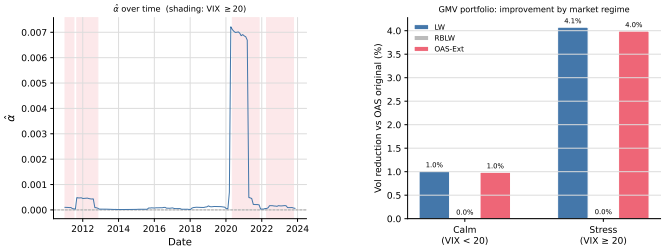


Fig. 2. Left: rolling-window $\hat{\alpha}$ for $p = 50$ large-cap US equities (January 2010 – December 2023), stress periods (VIX $\geq$ 20) shaded red. Right: percentage improvement in GMV realized volatility over OAS by market regime. OAS-Extended and LW deliver $\sim$1% in calm and $\sim$4% in stress periods; RBLW's improvement is economically negligible ($<$0.001% in both regimes).

VII. Conclusion

We have derived a closed-form correction to the Oracle Approximating Shrinkage estimator valid for any distribution with finite fourth moments. The correction enters through a single scalar α aggregating the fourth-order joint cumulants of the data; it vanishes under Gaussianity, recovering the OAS formula as a special case. A consistent $O(np)$ sample estimator $\hat{\alpha}$ is available in closed form, and the corrected shrinkage intensity lies in $[0, 1]$ exactly for both the population cumulant and its sample analogue.

Simulations and a rolling-window equity study confirm that the correction closes the gap between OAS and Ledoit–Wolf under heavy-tailed data, with gains largest at small n/p and in high-stress market regimes. The simulation study is limited to Student-t distributions and an AR(1) covariance structure, and the financial application covers a single universe of US large-cap equities; generalization to other settings remains to be assessed. The framework applies equally to neural signal covariance estimation, radar, and other high-dimensional settings where heavy-tailed observations are common. Natural extensions include applying the framework to structured shrinkage targets such as factor models or banded covariance matrices, and examining the behavior of $\hat{\alpha}$ when $n/p \ll 1$.

Appendix

Proof. Both $\mathrm{Var}(\|X\|^2)$ and $\widehat{\mathrm{Var}}(\|X\|^2)$ are non-negative (the latter by Jensen's inequality on the empirical distribution). Writing A for either Σ or $\hat{S}$ and V for the corresponding variance, since $\alpha = V - 2\,\mathrm{Tr}(A^2)$ we have $\alpha \geq -2\,\mathrm{Tr}(A^2)$. It then suffices to show $M \geq 2(1 - 1/p)\,\mathrm{Tr}(A^2)$, i.e., $\mathrm{Tr}^2(A) \geq \mathrm{Tr}(A^2)$. For any positive semidefinite A with eigenvalues $\lambda_i \geq 0$,

$$\mathrm{Tr}^2(A) - \mathrm{Tr}(A^2) = 2\sum_{i<j} \lambda_i \lambda_j \geq 0.$$

Hence $1 + \beta \geq 0$ and $\rho_{\text{OAS-E}} \in [0, 1]$. □

AI Use Disclosure

The author used Claude (Anthropic) to assist with prose editing, readability improvements, simulation code, and empirical analyses. All mathematical derivations and intellectual contributions are solely the author's own work.

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